



Programming assignment

Week 2 - Submitting Jobs With Slurm - Part 2

What to expect

 Due Sep 16, 11:59 PM IST Unlimited attempts[Instructions](#) [My submissions](#) [Discussions](#)

In this exercise, we are going to practice running parallel code that uses the Message Passing Interface MPI library. Since you haven't yet learned how to compile and run an actual MPI code, I will give you the source code and instructions on how to compile and run the parallel code, as the purpose of this exercise is to demonstrate you know how to submit a job for a parallel program!

In your Jupyter environment, there is a Fortran script, `hello_new.F90`, which prints "Hello World" on every core you run the program on. Additionally, the script will perform some basic calculations. To complete the exercise, you will need to write a batch script that:

1. Sets the number of nodes to 4
2. Sets the standard output to a file called, `week2_hellompi.txt`.
3. Sets the maximum run time to 30 seconds.
4. Unloads all loaded modules
5. Loads the `openmpi` or `mpich` module
6. Includes the following lines in the script. The first line compiles the `hello_new.f90` program. The second line executes the program on 4 processes.

```
1  mpif90 hello_new.f90 -o hello_new
2  mpirun -np 4 ./hello_new
```

Test your script using the terminal `sbatch` command. Once you are satisfied with your solution, submit your `week2_hellompi.txt` output file for grading. The autograder will evaluate:

- 4 lines starting with "Hello World from process",
- 4 lines containing exponential calculations (`Exp()`), and
- the number of mpi processes.

 [Report an issue](#)[Start your mini cluster. Work this week in "projects/week2-slurm2"](#)